

Lemma G.5 (Huang et al. (2023), Proposition 9). *Let $X_1, \dots, X_n$ be i.i.d. random vectors in $\mathbb{R}^p$, and let h be a function $h : \mathbb{R}^p \times \mathbb{R}^p \rightarrow \mathbb{R}$ satisfying $\mathbf{E}(h(X_1, X_2)|X_1) = 0$. Suppose there is a sequence of functions $\phi_1, \dots, \phi_K : \mathbb{R}^p \rightarrow \mathbb{R}$ such that h can be represented as*

$$h(x, x') = \sum_{k=1}^K \phi_k(x) \phi_k(x')$$

for all $x, x' \in \mathbb{R}^p$. Also, let $\phi(X_1) = (\phi_1(X_1), \dots, \phi_K(X_1))$ and let $\Sigma \in \mathbb{R}^{K \times K}$ denote the covariance matrix of $\phi(X_1)$. Lastly, let $\tau^2 = \text{var}(h(X_1, X_2))$, and let $Z_1, \dots, Z_K$ denote independent standard normal random variables. Then,

$$\sup_{t \in \mathbb{R}} \left| \mathbf{P} \left(\sum_{1 \leq i \neq j \leq n} \frac{h(X_i, X_j)}{\sqrt{\tau^2 n(n-1)}} \leq t \right) - \mathbf{P} \left(\sum_{k=1}^K \frac{1}{\tau} \lambda_k(\Sigma) (Z_k^2 - 1) \leq t \right) \right| \lesssim n^{-\frac{1}{5}} + n^{-\frac{1}{14}} \left(\frac{1}{\tau} \|h(X_1, X_2)\|_{L^3} \right)^{\frac{3}{7}}.$$